Partial Retinal Image Reconstruction with CVAE

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1. Introduction

Optical coherence tomography angiography (OCTA) is a noninvasive imaging technique that visualizes retinal microvasculature and supports quantitative assessment of perfusion-related biomarkers that are useful in routine ophthalmology and research workflows [1]. OCTA has become especially relevant for diseases where microvascular disruption is central to diagnosis and monitoring, including diabetic retinopathy (DR) and age-related macular degeneration (AMD), because it can reveal capillary-level changes without dye injection and can be repeated frequently for longitudinal follow-up [2] [3]. At the same time, OCTA quality is highly sensitive to practical acquisition constraints such as motion, noise, and limited sampling, which can degrade visibility of small vessels and boundary-sensitive regions like the FAZ. These issues directly impact clinical interpretability and the reliability of downstream measurements [1] [2].

This project focuses on reconstructing high-quality OCT/OCTA volumes from incomplete measurements, motivated by the idea that faster or more efficient acquisitions can reduce scan time and motion artifacts but turn reconstruction into an ill-posed inverse problem [4]. In this setting, the algorithm must not only recover global structure, but also preserve fine, clinically meaningful vascular detail that is easy to blur, distort, or hallucinate when information is missing. Recent work has shown that deep learning can act as a powerful learned prior for reconstruction in optical coherence tomography contexts, enabling more faithful recovery from constrained acquisitions compared to traditional pipelines [5]. Building on this direction, we use a **3D Conditional Variational Autoencoder (CVAE)** as the main architecture in this paper, explicitly conditioning the reconstruction on the masked observation and its sampling mask [6]. For comparison, we also train a standard 3D VAE as a baseline and include a Vector-Quantized VAE (VQ-VAE) as an additional reference model with a discrete latent space that can promote sharper structure [7].

1.1. Project Goal

The objective of this project is to reconstruct high-quality 3D OCTA patches from incomplete measurements generated by random voxel masking (sampling rate 0.6). Our primary focus is a 3D Conditional Variational Autoencoder that conditions the reconstruction on the masked observation and its binary sampling mask, and we compare it against a standard 3D VAE baseline and a VQ-VAE reference model.

Overall, the goal was achieved. The CVAE learned to reliably recover the missing regions and produced reconstructions that are close to the ground truth in both voxel-wise fidelity and structural similarity (e.g., test PSNR $\approx$ 39.1 dB and SSIM $\approx$ 0.981), substantially improving over the baseline VAE. While the VQ-VAE produced slightly sharper outputs and higher PSNR, the CVAE provided the best overall balance and the most consistent reconstructions for the masked-input setting used in this work.

1.2. Dataset Description

In this project we use only the OCTA-6mm portion of the OCTA-500 dataset, which includes 300 subjects (one randomly selected eye per subject for independence) collected at Jiangsu Province Hospital between March 2018 and July 2020 using an RTVue-XR (Optovue) 70 kHz spectral-domain OCT

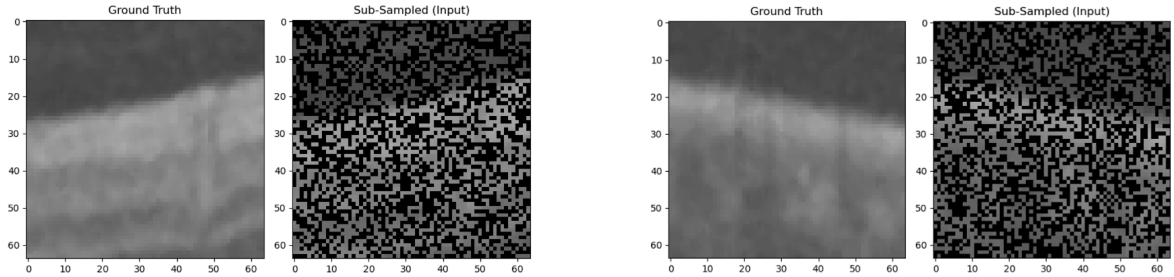
device [4]. Each case provides paired 3D OCT and 3D OCTA macular volumes covering 6 mm × 6 mm × 2 mm, with a volumetric resolution of 400 × 400 × 640 pixels for OCTA-6mm [4]. Although the raw data are volumetric, the dataset is also released as 2D B-scan images for easier use; for OCTA-6mm this corresponds to 240,000 2D scans (300 subjects × 400 B-scans per volume × 2 modalities: OCT and OCTA) derived from the volume dimensions and the dataset's 2D-scan formatting [4]. In terms of class distribution, OCTA-6mm is heavily skewed toward pathology: the normal group is a minority and approximately 69.7% of subjects are diagnosed with retinal disease, including common conditions such as AMD, DR, CNV, CSC, and RVO (plus a small "other" category) [4].

1.3. Data Processing

For data processing, we first reorganized the raw OCTA-6mm data at the subject-volume level by loading each subject's ordered 2D scans and reassembling them into a full 3D volume, then splitting each volume into smaller 3D patches of size 64×64×64 with overlap (stride 48, overlap 16) and saving each patch as a compressed file; during this patch extraction step we also applied a median filter with kernel size 3 to suppress speckle-like noise. After generating these patch files, we created non-overlapping train/validation/test splits by shuffling subject folders with a fixed random seed (42) and partitioning them into 90% training, 5% validation, and 5% test, where all patches and slices coming from the same subject volume were kept inside the same split (files are collected recursively only within the folders assigned to each split). This design is the main protection against data leakage, because it prevents patches derived from the same original 3D acquisition (highly correlated in space and across adjacent slices) from being scattered across training and evaluation sets. At training time, each sample is loaded, converted to floating-point, and formatted with an explicit channel dimension; we do not resize data, preserving the native sampling of the stored patches. Finally, to model incomplete acquisition, we generate a random Bernoulli sampling mask and apply it to create a sub-sampled input, using a sampling rate of 0.6 (60% of voxels observed on average), and we concatenate the mask as an additional channel so the model can explicitly condition on which regions were observed versus missing.

1.4. Input and Ground Truth

After preprocessing, each training sample is a 3D OCTA patch of size 64×64×64 extracted from the reconstructed subject volumes (with overlap) and denoised using a 3×3×3 median filter. The final model input is a sub-sampled version of this patch, created by applying a random Bernoulli sampling mask with sampling rate 0.6, meaning roughly 60% of voxels are retained while the rest are set to zero (as illustrated in the "Sub-Sampled (Input)" examples in Figure 1). To make the reconstruction problem well-posed for the conditional models, we also provide the sampling mask itself as an additional channel, so the network explicitly knows which locations were observed versus missing. The ground truth is the corresponding fully observed patch before masking (as seen in figure 1), used as the reconstruction target under a pixel-wise loss. This setup matches the proposal stage exactly: we reconstruct missing retinal structure from incomplete measurements using masked inputs paired with their original unmasked targets, with no changes in the definition of inputs or ground-truth labels.



(a) Sample 1 (ground truth vs. input) (b) Sample 2 (ground truth vs. input)

Figure 1: Representative OCTA patches. For each sample, the left panel shows the ground truth patch and the right panel shows the corresponding sub-sampled input produced with a random Bernoulli mask at sampling rate 0.6.

2. Methodology

This section describes the modeling approach used to reconstruct 3D OCTA patches from incomplete measurements generated via random voxel masking. Our primary contribution and main model is a 3D Conditional Variational Autoencoder (CVAE), which explicitly conditions the reconstruction on the masked observation and its binary sampling mask. For context and fair evaluation, we also describe two additional generative models, a standard 3D VAE and a vector-quantized VAE (VQ-VAE). These architectures are later compared under the same dataset protocol and evaluation metrics.

2.1. Baseline model: Variational Autoencoder

As a baseline, we implement a 3D Variational Autoencoder (VAE) for volumetric OCTA reconstruction. The input to the network is a two-channel volume consisting of the masked observation and its binary sampling mask. The encoder uses a small hierarchy of 3D convolutional layers (kernel size $3 \times 3 \times 3$) with downsampling to compress the input into a compact representation, which is then projected to the mean and log-variance of a diagonal Gaussian latent variable z. Sampling is performed with the reparameterization trick,

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

to enable end-to-end training. The decoder mirrors the encoder by expanding z through a linear projection and upsampling back to the original voxel resolution using 3D transposed convolutions (stride 2), producing a reconstructed volume $\hat{x}$ followed by a sigmoid activation to bound intensities to $[0, 1]$.

Training optimizes a β-VAE objective that balances voxel-wise fidelity and latent regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \beta\, D_{\text{KL}}(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)) \;\; (+\, \lambda\, \mathcal{L}_{\text{perc}}),$$

where $\mathcal{L}_{\text{MSE}}$ is the reconstruction loss and the KL term discourages posterior collapse; an optional perceptual term $\mathcal{L}_{\text{perc}}$ can be included to better preserve high-frequency structure.

2.2. Conditional Variational Autoencoder

To better handle reconstruction from incomplete OCTA measurements, we use a **3D Conditional Variational Autoencoder (CVAE)** that models the conditional generative process $p_\theta(x \mid z, c)$, where $x \in \mathbb{R}^{1 \times D \times H \times W}$ is the ground-truth OCTA patch and c is the conditioning tensor formed by the masked observation and its binary sampling mask (two channels). The encoder defines a posterior $q_\phi(z \mid x, c)$ by concatenating x and c channel-wise and passing them through a **fully convolutional 3D backbone** with **three downsampling stages**; each stage is a residual-style block composed of **two 3D convolutions with** $3 \times 3 \times 3$ **kernels**, normalization (GroupNorm), and a smooth nonlinearity, with the first convolution in each stage using **stride 2** to reduce spatial resolution. The resulting feature volume is compressed to a fixed-length vector using **adaptive global average pooling**, and

two linear projections output the mean and log-variance of a diagonal Gaussian latent distribution. The conditional prior $p_\psi(z \mid c)$ is implemented with a second encoder of the same structure that consumes only c, producing $(\mu_p(c), \log \sigma_p^2(c))$. The latent sample is obtained with the standard reparameterization trick,

$$z = \mu_q(x, c) + \sigma_q(x, c) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

which enables end-to-end optimization. The decoder then reconstructs a full patch using a conditional decoder $p_\theta(x \mid z, c)$: z is concatenated with a compact conditioning vector extracted from c, mapped by fully connected layers into an initial $2 \times 2 \times 2$ 3D feature grid, and progressively upsampled with **trilinear interpolation** until the target patch size is reached. At each upsampling scale, the decoder injects **spatial conditioning** by resizing the masked observation and mask to the current resolution (continuous channels via trilinear interpolation and the binary mask via nearest-neighbor) and concatenating them with the decoder feature maps, followed by additional $3 \times 3 \times 3$ convolutional refinement blocks. Channel width is gradually reduced near the output through successive convolutional blocks, ending with a final $3 \times 3 \times 3$ convolution that predicts the reconstructed OCTA intensity volume. Training maximizes the conditional evidence lower bound (ELBO),

$$\mathcal{L}_{\text{CVAE}} = \mathbb{E}_{q_\phi(z|x,c)}\big[\log p_\theta(x \mid z, c)\big] - \beta\, D_{\text{KL}}(q_\phi(z \mid x, c) \,\|\, p_\psi(z \mid c)),$$

where the reconstruction term is computed voxel-wise (MSE or L_1), and the KL term uses the closed-form divergence between diagonal Gaussians,

$$D_{\text{KL}}(q\|p) = \frac{1}{2} \sum_i \left(\log \frac{\sigma_{p,i}^2}{\sigma_{q,i}^2} + \frac{\sigma_{q,i}^2 + (\mu_{q,i} - \mu_{p,i})^2}{\sigma_{p,i}^2} - 1 \right).$$

Finally, because the condition explicitly includes the sampling mask, the model enforces **data consistency** at observed voxels by replacing the reconstructed values with the measured ones wherever the mask indicates valid measurements, ensuring the network focuses its capacity on inferring only the missing regions.

2.3. Vector Quantized Variational Autoencoder

As an additional comparison, we consider a volumetric Vector Quantized VAE (VQ-VAE) designed for reconstructing 3D OCTA volumes from sparsely sensed inputs (random voxel dropout). Instead of relying on max-pooling or strided convolutions for downsampling, the encoder performs a 3D Discrete Haar Wavelet Transform (DWT), which decomposes the feature volume into eight sub-bands (one low-frequency component, LLL, and seven high-frequency components) to better preserve edge and vessel details.

Wavelet decomposition is applied across multiple levels; each set of DWT coefficients is processed by lightweight "mini" U-Nets that extract features without changing coefficient resolution. To form a single latent representation, lower-level coefficients are downsampled to match the deepest level resolution and concatenated, then passed through a final aggregation block to produce a compact latent tensor (e.g., $z \in \mathbb{R}^{16\times8\times8\times8}$).

The continuous latent space is replaced by a deterministic vector quantizer with a learnable codebook $E = \{e_1, \ldots, e_K\}$, $e_k \in \mathbb{R}^{D_{\text{emb}}}$. Each latent vector is mapped to its nearest codebook entry:

$$z_q = e_k, \qquad k = \arg\min_j \|z - e_j\|_2.$$

The decoder then upsamples the quantized latent through a small stack of residual blocks to recover the original voxel resolution (e.g., $8^3 \to 16^3 \to 32^3 \to 64^3$), followed by a final 3D convolution to output the reconstructed volume $\hat{x}$.

Training uses the standard VQ-VAE objective, combining a reconstruction term with codebook and commitment losses:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \|\text{sg}[z] - z_q\|_2^2 + \beta\|z - \text{sg}[z_q]\|_2^2,$$

where sg[·] denotes the stop-gradient operator.

2.4. Training Details and Reproducibility

All CVAE experiments were implemented using PyTorch. Training used the **Adam** optimizer and we performed a small learning-rate sweep over 1×10^{-3}, 3×10^{-3}, and 5×10^{-3} (comparative results are reported in Table 3). The model was optimized with the same objective defined in Section 2.2, $\mathcal{L}_{\text{CVAE}}$, combining a voxel-wise reconstruction term with a KL regularizer weighted by β; in our runs $\beta = 0.5$ with an optional linear warm-up over the first 5 epochs to stabilize early training. We trained for 10 epochs using a batch size of 8 for training and 4 for validation, reflecting the memory cost of 3D convolutions on 64^3 patches. For reproducibility, we fixed the data split seed to 42 so the subject-level 90/5/5 train/validation/test partition is deterministic across runs. All training was executed on the **GAIVI** cluster using an **NVIDIA A100 GPU**, and we enabled mixed-precision training (automatic casting plus gradient scaling) to improve throughput on CUDA hardware.

3. Performance Improvement Techniques

(1) Data augmentation via stochastic sub-sampling masks. To make the CVAE robust to different missing-data patterns, we used implicit augmentation by generating a new random Bernoulli sampling mask for every training sample and iteration, then masking the OCTA patch to create the conditional input (masked observation + mask). This means the same underlying ground-truth patch can appear with many different observation patterns over training, which regularizes the model and improves generalization to unseen masks at test time. In practice, this augmentation reduced overfitting (smaller train–validation gap) and produced more stable validation PSNR/SSIM compared to training on a fixed mask pattern.

(2) Dropout regularization (rates: 0, 0.05, 0.1). We inserted 3D dropout after convolutional blocks (and once after the decoder's initial fully connected projection) so that intermediate feature activations are randomly suppressed during training. Conceptually, dropout is applied as

$$\tilde{h} = \frac{m \odot h}{1-p}, \qquad m \sim \text{Bernoulli}(1-p),$$

where p is the dropout rate and h is a feature tensor. As shown in Table 1, with $p = 0.1$, we obtained the strongest validation performance among the tested dropout rates (loss 1.851E−04, PSNR 39.372 dB, SSIM 0.9811), and it also achieved the highest test SSIM (0.9812). The intermediate value $p = 0.05$ was close but consistently worse on validation (loss 1.865E−04, PSNR 39.302 dB, SSIM 0.9808) and did not improve test error relative to $p = 0.1$. Notably, $p = 0$ yielded a slightly lower test loss and higher test PSNR (loss 1.89E−04, PSNR 39.247 dB), but its validation metrics were weaker and its test SSIM was lower (0.9800). Since our model selection is based on validation performance, we use $p = 0.1$ in the remaining experiments.

Table 1: Performance Comparison Across Dropout Rates

	Validation (Best Epoch)			**Test Performance**		
Dropout Rate	**Loss**	**PSNR (dB)**	**SSIM**	**Loss**	**PSNR (dB)**	**SSIM**
0	1.965E-04	39.282	0.9802	1.89E-04	39.247	0.9800
0.05	1.865E-04	39.302	0.9808	1.96E-04	39.137	0.9802
0.10	**1.851E-04**	**39.372**	**0.9811**	**1.90E-04**	**39.106**	**0.9812**

(3) Normalization choice: Group Normalization vs Batch Normalization. Each 3D convolutional block includes a normalization layer. Batch Normalization (BN) normalizes per-channel activations using batch statistics,

$$\text{BN}(h) = \gamma \frac{h - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta,$$

while Group Normalization (GN) computes statistics over groups of channels within each sample (independent of batch size),

$$\text{GN}(h) = \gamma \frac{h - \mu_G}{\sqrt{\sigma_G^2 + \epsilon}} + \beta.$$

As shown in Table 2, GN consistently worked better than BN because training uses small 3D batches (memory-limited), so BN's batch mean/variance estimates are noisy and can fluctuate strongly from step to step, destabilizing optimization and hurting validation performance. GN avoids that issue by not relying on batch-level statistics, which makes it much more reliable for 3D convolutional models under tight GPU memory constraints. In our runs, switching from BN to GN produced smoother loss curves and better reconstruction quality on validation samples.

Table 2: Performance Comparison Across Normalization Methods

	Validation (Best Epoch)			**Test Performance**		
Normalization	**Loss**	**PSNR (dB)**	**SSIM**	**Loss**	**PSNR (dB)**	**SSIM**
Group	**1.851E-04**	**39.372**	**0.9811**	**1.90E-04**	**39.106**	**0.9812**
Batch	1.923E-04	38.804	0.9756	1.98E-04	38.547	0.9723

4. Performance Evaluation

We evaluate reconstruction quality using **PSNR** and **3D SSIM**, computed between the reconstructed OCTA patch $\hat{x}$ and the ground-truth patch x (both scaled to $[0, 1]$ via per-patch min–max normalization during evaluation). **Peak Signal-to-Noise Ratio (PSNR)** summarizes voxel-wise fidelity through the mean squared error (MSE):

$$\mathrm{PSNR}(x, \hat{x}) = 10 \log_{10}\left(\frac{L^2}{\mathrm{MSE}(x, \hat{x})}\right), \qquad \mathrm{MSE}(x, \hat{x}) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2,$$

where L is the data range (here $L = 1$) and N is the number of voxels. Higher is better, and PSNR is unbounded above in theory (it approaches ∞ for a perfect reconstruction), but in practice it is reported in dB and commonly falls in a finite range for real reconstructions. **3D Structural Similarity (SSIM-3D)** captures structural agreement (luminance, contrast, and local texture) using local statistics computed over a Gaussian window:

$$\mathrm{SSIM}(x, \hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)},$$

where μ and σ^2 are local means and variances, $\sigma_{x\hat{x}}$ is local covariance, and C_1, C_2 are stability constants. SSIM is typically reported in the range $[0, 1]$ for normalized images, where 1 indicates perfect similarity and higher is better. For reporting, both metrics are computed per sample and then averaged across the validation/test set to produce a single PSNR and SSIM score per experiment.

Quantitative Results. To identify the optimal optimization setting, we performed a learning-rate ablation with the dropout rate fixed at 0.1.

As shown in Table 3, the learning rate 1e−3 achieved the best validation performance (lowest validation loss 1.851E−04, highest validation PSNR 39.372 dB, and highest validation SSIM 0.9811) and was therefore selected for subsequent experiments. On the test set, this setting remains consistently strong (test loss 1.90E−04, PSNR 39.106 dB, SSIM 0.9812).

Table 3: Performance Comparison Across Learning Rates

	Validation (Best Epoch)			**Test Performance**		
Learning Rate	**Loss**	**PSNR (dB)**	**SSIM**	**Loss**	**PSNR (dB)**	**SSIM**
1.00E-03	**1.851E-04**	**39.372**	**0.9811**	**1.90E-04**	**39.106**	**0.9812**
3.00E-03	1.908E-04	39.221	0.9798	1.94E-04	39.148	0.9811
5.00E-03	1.911E-04	39.104	0.9793	1.97E-04	39.048	0.9794

The learning curves in Figure 2 show that both training and validation loss decrease rapidly in the first epochs and then gradually stabilize, with a relatively small gap between curves, suggesting limited

overfitting under the selected hyperparameters. Complementing this, Figure 3(a) and Figure 3(b) track reconstruction quality directly: PSNR and SSIM increase steadily across epochs and then plateau, indicating that optimization improves both voxel-wise fidelity (PSNR) and structural similarity (SSIM) until convergence.

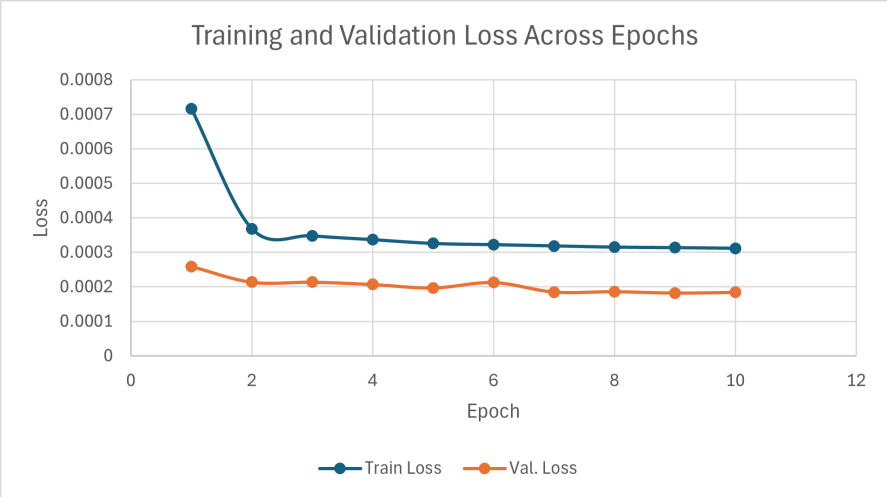
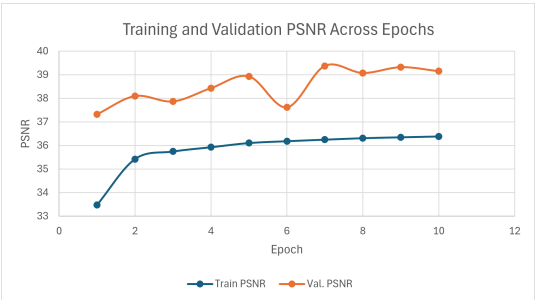
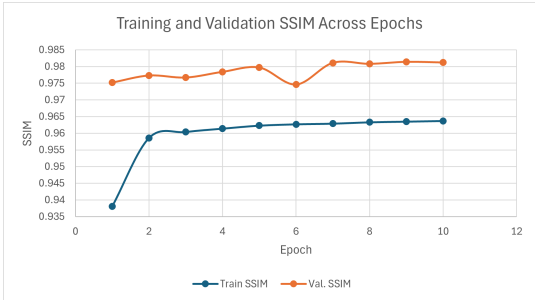


Figure 2: Training and validation loss across epochs for the CVAE model.



(a) CVAE PSNR Across Epochs



(b) CVAE SSIM Across Epochs

Figure 3: Training dynamics of the CVAE model showing PSNR and SSIM evolution across epochs.

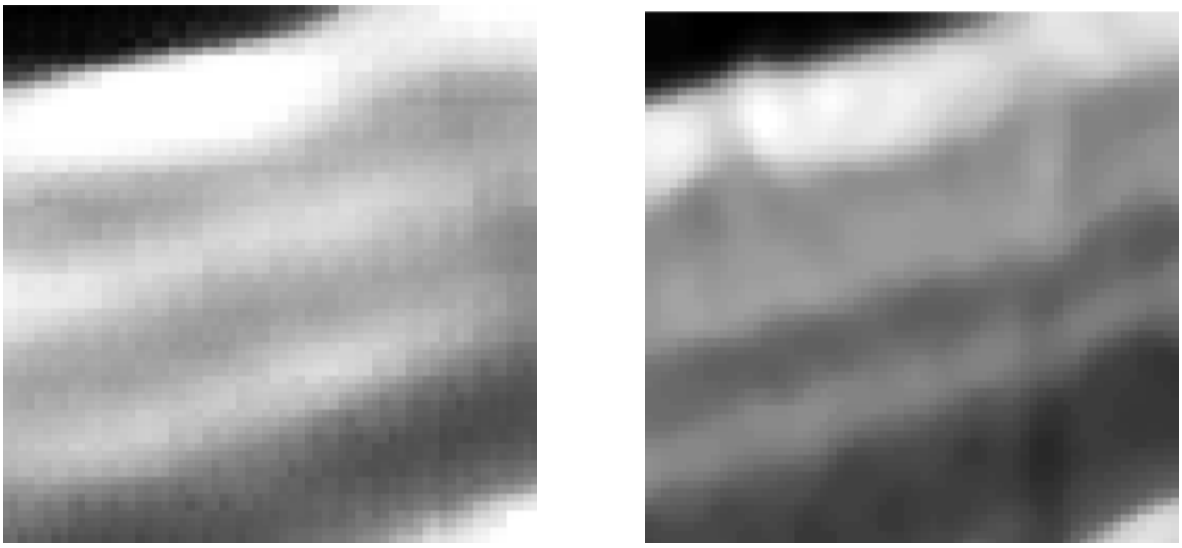
5. Comparison of Approaches

This section compares the baseline VAE against the two stronger generative architectures (CVAE and VQ-VAE) under the same dataset split, masking protocol (sampling rate 0.6), and evaluation metrics (PSNR and SSIM-3D). Table 4 summarizes the quantitative performance across models. The standard VAE performs substantially worse (test PSNR 30.605 dB, SSIM 0.844), which is expected because it does not explicitly model the missing-data pattern beyond the input itself and tends to produce overly smoothed reconstructions. In contrast, the CVAE improves reconstruction fidelity dramatically (test PSNR 39.106 dB, SSIM 0.9812) by conditioning the latent distribution and the decoder on the masked observation and the binary sampling mask, which makes the inverse problem better posed and encourages data-consistent completions. The VQ-VAE achieves the highest PSNR (test PSNR 42.631 dB) and often produces sharper-looking structure due to its wavelet-based multiscale representation and discrete codebook, although its test SSIM (0.9202) suggests that these sharper reconstructions can introduce local structural deviations that SSIM penalizes. Overall, the results indicate that conditioning (CVAE) is a major driver of performance gains over the baseline, while discrete latents and multiscale decomposition (VQ-VAE) further improve voxel-wise fidelity and perceived sharpness.

Table 4: Performance Comparison Across Generative Models

		Validation (Best Epoch)		**Test Performance**	
Model	**Training Epochs**	**PSNR (dB)**	**SSIM**	**PSNR (dB)**	**SSIM**
VAE	10	28.589	0.838	30.605	0.844
CVAE	10	39.372	0.981	39.106	0.981
VQ-VAE	20	43.476	0.982	42.631	0.920

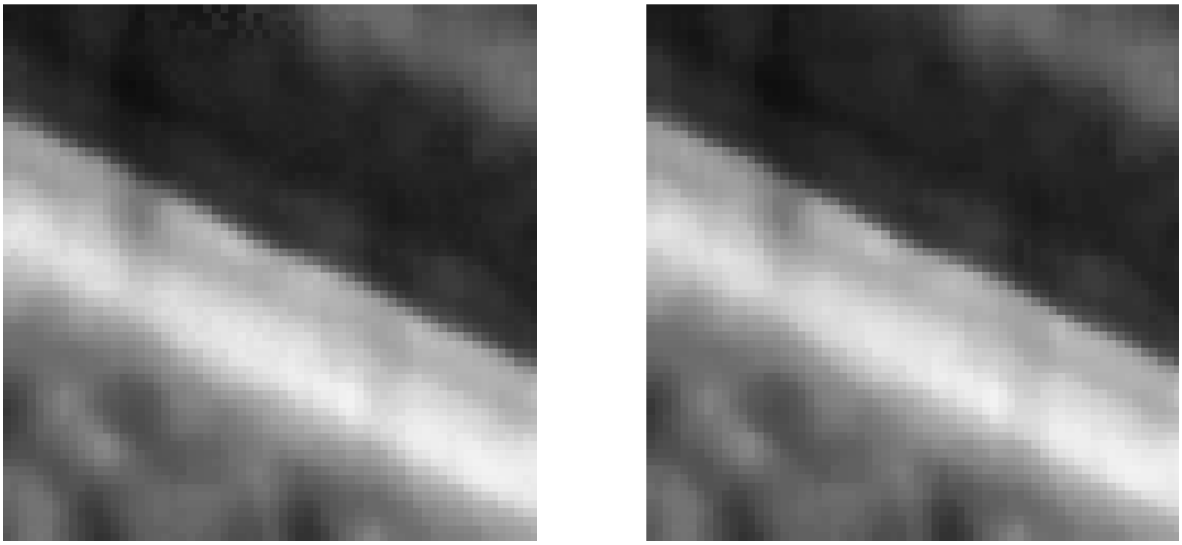
Figure 4 shows a representative VAE reconstruction compared to its ground truth. The reconstruction captures the coarse intensity distribution but loses fine boundary definition and exhibits visible blurring, especially along elongated bright bands and transition regions. This aligns with the low PSNR and SSIM in Table 4 and highlights the baseline model's tendency to average over uncertainty when large portions of the input are missing.



(a) VAE Reconstruction (b) VAE Ground Truth

Figure 4: Comparison between VAE reconstructed output and ground truth sample.

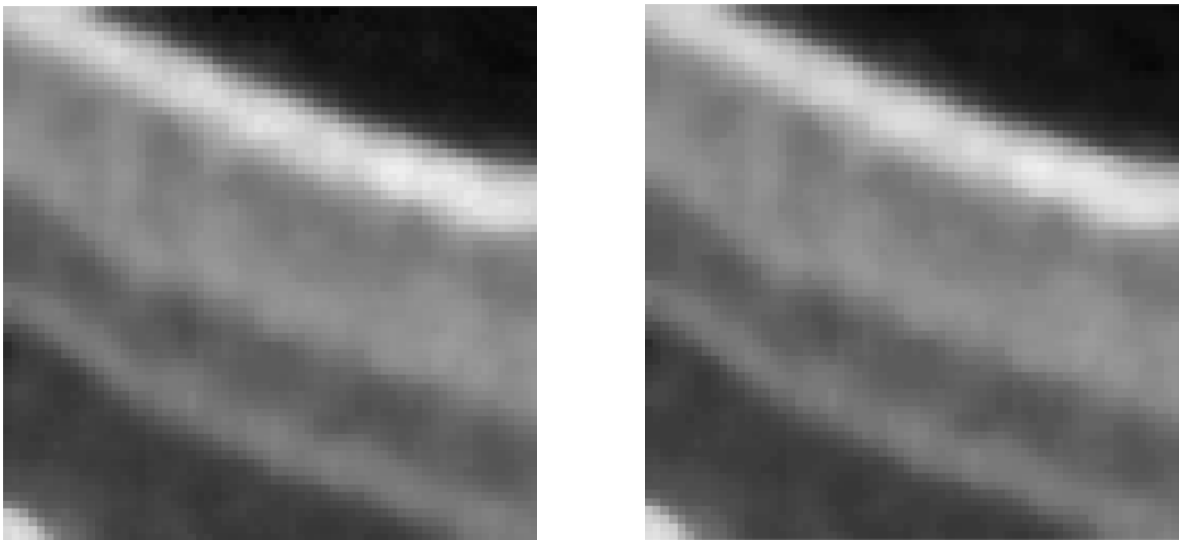
Figure 5 demonstrates that the CVAE produces reconstructions that are much closer to the ground truth in both global structure and local contrast. The dominant layered patterns are recovered with good continuity, and the overall appearance is clinically more plausible than the baseline. Minor differences remain, mainly as slight smoothing or small spatial shifts at sharp boundaries and subtle intensity mismatches in darker regions, but these are comparatively small relative to the baseline error and are consistent with the high SSIM reported for CVAE.



(a) CVAE Reconstruction (b) CVAE Ground Truth

Figure 5: Comparison between CVAE reconstructed output and ground truth sample.

Figure 6 shows a VQ-VAE reconstruction versus ground truth. Visually, the reconstruction appears slightly sharper than the CVAE, with clearer separation of bright bands and improved edge localization. This qualitative improvement matches the higher PSNR in Table 4 and is consistent with the VQ-VAE design, which combines multiscale wavelet decomposition with a discrete latent codebook that can reduce over-smoothing. At the same time, the lower test SSIM suggests that some of these sharper details may come with small local structural distortions that affect SSIM more strongly than PSNR.



(a) VQ-VAE Reconstruction (b) VQ-VAE Ground Truth

Figure 6: Comparison between VQ-VAE reconstructed output and ground truth sample.

Across all experiments, the baseline VAE is the weakest approach and struggles to preserve structure under sparse sampling. The CVAE provides the best overall balance, delivering consistently strong quantitative performance and high-quality reconstructions by explicitly conditioning on the sampling mask and enforcing data consistency at observed voxels. The VQ-VAE is slightly better in terms of voxel-wise fidelity and perceived sharpness (highest PSNR), likely due to its multiscale wavelet features and discrete latent representation, although it shows a trade-off in SSIM on the test set. Taken together,

these results support using the CVAE as the primary model for reliable reconstructions, with the VQ-VAE as the strongest comparison when prioritizing sharper outputs and higher PSNR.

6. Foundation Model Comparison

The rubric includes a "foundation model" comparison (e.g., CLIP) against a CNN-based approach. In this project, however, our reconstruction pipeline is based on probabilistic generative models (VAE, CVAE, and VQ-VAE) operating directly on 3D OCTA volumes, and we did not incorporate a pretrained foundation model component. In particular, models such as CLIP are primarily designed for aligned image–text representation learning and retrieval, and they do not provide a direct, drop-in solution for supervised 3D voxel reconstruction from sparse measurements under our experimental setup.

For this reason, and consistent with the professor's guidance in class that a foundation-model baseline is not required when the project's main approaches are VAE-based generative reconstructions, we omit this comparison and instead focus on a fair evaluation among the three implemented generative architectures (baseline VAE, main CVAE, and VQ-VAE) under the same dataset protocol and metrics.

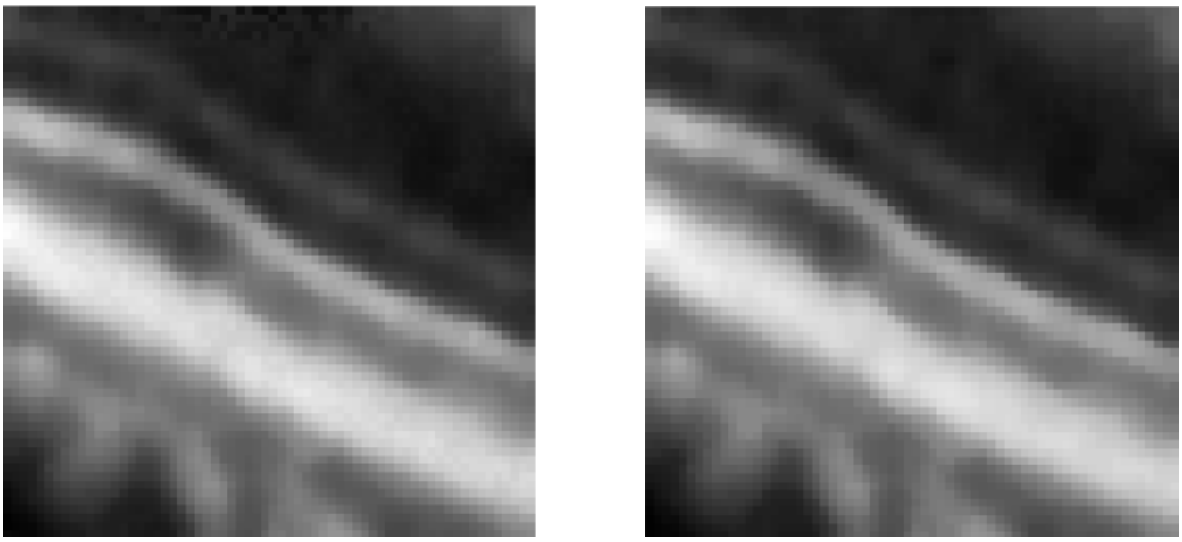
7. Error and Failure Analysis

7.1. Common failure cases

Across the qualitative examples, the most consistent failure modes for the CVAE are: (i) **imprecise recovery of linear or edge-like structures**, where elongated bright bands or layer boundaries are reconstructed with slight angular deviations or spatial shifts; and (ii) **imperfect intensity matching**, particularly in darker regions where the reconstructed black tones do not align with the ground truth, leading to mild contrast and shading errors.

7.2. Qualitative examples of errors

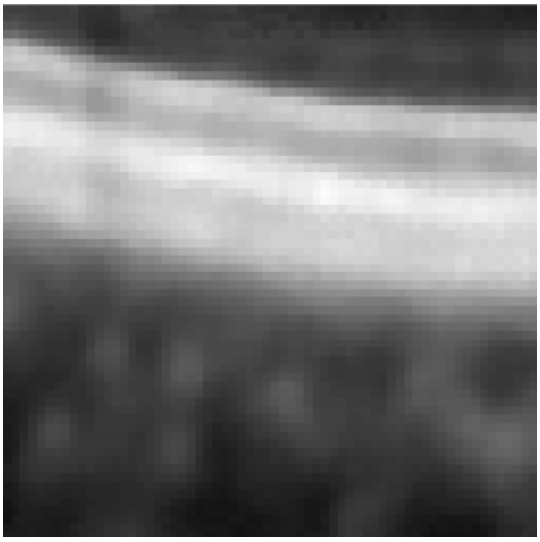
In Figure 7, the reconstruction captures the global banded appearance but struggles to preserve the exact geometry of prominent linear tendencies. In particular, some boundary-like lines appear slightly rotated or "rounded" compared to the ground truth, consistent with edge smoothing and small angular drift. Additionally, several darker areas in the reconstruction do not reach the same low-intensity levels as the ground truth, suggesting a bias toward mid-range intensities rather than precise dark-tone recovery.



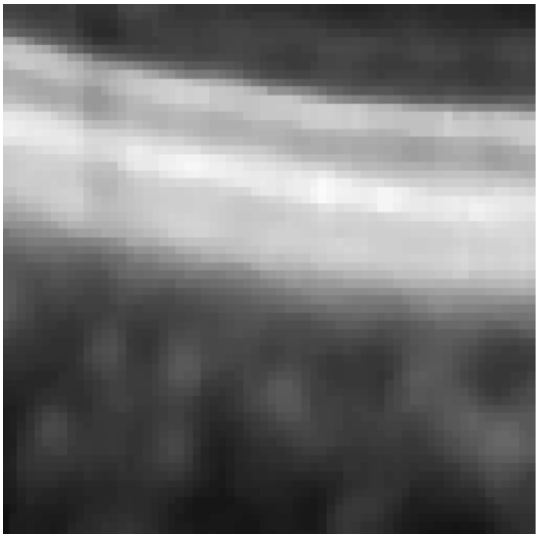
(a) Reconstruction 1 (b) Ground Truth 1

Figure 7: Comparison between reconstructed output and ground truth sample 1 (CVAE).

Figure 8 shows a similar pattern: the dominant bright layer is recovered, but its boundaries are smoothed and less sharply localized than in the ground truth. The reconstructed image also exhibits subtle mismatches in dark regions, where the background texture and black tones are not reproduced with the same depth, resulting in mild contrast compression.



(a) Reconstruction 2



(b) Ground Truth 2

Figure 8: Comparison between reconstructed output and ground truth sample 2 (CVAE).

7.3. Observed patterns and implications

Two patterns stand out. First, **structure errors are concentrated around high-frequency content** (edges, thin lines, and sharp transitions), where the model tends to produce smoother, more averaged boundaries. Second, **intensity errors are more visible in low-signal areas**, where the model appears conservative and underestimates extreme dark values. These trends are typical of generative reconstruction trained with voxel-wise objectives, where averaging over uncertain missing regions can yield visually plausible but slightly blurred outputs. In practice, this means that while the model is reliable for global structure recovery, it is less reliable for fine-grained geometric fidelity (exact line angles) and for accurately restoring contrast in dark regions, both of which are clinically relevant in OCTA-like imagery.

8. Team Collaboration

We organized the project around a shared experimental pipeline while developing the three architectures in parallel. As a team, we aligned early on the most appropriate dataset to use and agreed on a consistent data splitting strategy (train/validation/test) to ensure fair and comparable evaluations across models. We also collaborated on making the dataset suitable for training by discussing preprocessing choices and input/target formatting so that each architecture could be trained under consistent assumptions.

During implementation, we maintained frequent check-ins to compare design decisions and debug issues. We helped each other by reviewing how each architecture was implemented, sharing what training techniques improved stability and reconstruction quality, and discussing trade-offs observed during experimentation. Collectively, this collaboration resulted in three working generative models trained and evaluated under a unified dataset and evaluation framework, enabling meaningful comparison across approaches.

9. Individual Contributions

Lorenzo: Implemented the Variational Autoencoder (VAE) architecture and conducted the corresponding training and evaluation runs within the shared dataset pipeline.

Samuel (me): Implemented the Conditional Variational Autoencoder (CVAE), including conditioning design decisions, training setup, and evaluation under the common preprocessing and split strategy.

Raniero: Implemented the Vector-Quantized VAE (VQ-VAE) architecture and carried out the associated experiments and evaluations using the same dataset protocol.

10. Conclusion

In this project, we studied probabilistic generative reconstruction of 3D OCTA patches from incomplete measurements generated by random voxel masking (sampling rate 0.6). Using OCTA-6mm volumes from OCTA-500, we implemented three generative models under a shared preprocessing pipeline and subject-level train/validation/test splits to avoid leakage. Our main architecture, a 3D Conditional Variational Autoencoder (CVAE), explicitly conditions reconstruction on the masked observation and its binary sampling mask and enforces data consistency at observed voxels.

Quantitatively, the CVAE achieved strong and stable performance (test PSNR 39.106 dB, SSIM 0.9812), substantially outperforming the baseline VAE (test PSNR 30.605 dB, SSIM 0.844). The VQ-VAE obtained the highest voxel-wise fidelity (test PSNR 42.631 dB) and produced slightly sharper qualitative reconstructions, although it showed a lower test SSIM (0.9202), indicating a trade-off between sharpness and local structural agreement as measured by SSIM. Overall, conditioning on the sampling mask was the most important factor for improving reconstruction quality over the baseline, while discrete latent representations and multiscale decomposition further improved perceived sharpness and PSNR.

Several practical lessons emerged. First, for 3D convolutional models with small batch sizes, Group Normalization was more reliable than Batch Normalization, producing better validation performance. Second, implicit augmentation through stochastic re-sampling masks each iteration improved robustness to different missing-data patterns. Finally, the qualitative error analysis suggests the CVAE's remaining failures are concentrated in high-frequency structures (thin edges and sharp transitions) and in low-signal dark regions, where reconstructions can become slightly smoothed or conservatively biased toward mid-range intensities. Future work can improve both fidelity and sharpness in several ways. A direct extension is to incorporate stronger structure-aware objectives (e.g., edge-weighted losses or multi-scale reconstruction terms) to reduce smoothing on thin vessel-like features, while keeping the CVAE's mask-conditioning and data-consistency constraint. Another direction is to evaluate multiple sampling rates and structured masks (rather than purely random Bernoulli) to better match realistic acquisition constraints. Finally, the best elements of both strong models could be combined by integrating discrete or multiscale latent representations into the conditional framework, aiming to preserve the CVAE's high structural consistency while achieving the sharper reconstructions observed with VQ-VAE.

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